

# CV

## Basics

---

**Name** Luke Lowery  
**Email** lukel@tamu.edu  
**Website** <https://lukelowry.github.io>  
**Summary** Currently employed at Texas A&M University. EE PhD Student of Power Systems

## Work

---

2024 - PRESENT

### **Research Assistant**

Texas A&M University

- Develops Phasor & EMT models to accelerate the integration of renewables in modern transmission systems.
- Performs novel research on the impact of Geomagnetic Disturbances on the nations electric grid.
- Applies modern mathematics to solve open problems in energy and power research.

2023.05 - 2023.08

### **High Voltage Intern**

SpaceX

- Switchgear & transformer installation, design, and coordination for mission-critical loads (5kV & 35kV)
- Responsible for differential and over-current protection on critical feeder lines to production and launch-pad.
- Designed & implemented 500 kW feeder and building distribution to a new production building in 4 weeks.
- Co-designed the microgrid controller functionality with Tesla Engineers. Functionality allowed our Mega-Packs (smart battery storage) to create seamless transitions from the Utility to isolated island operation.

2022.08 - 2023.01

### **Power Systems Cooperative Education**

SpaceX

- Coordinated with other SpaceX production and launch sites to create a standardized SCADA system. Performed transient analysis of synchronization anomalies to discover coordination errors.
- Introduced solar panel expansion to decrease diesel use, turning Starbase into a 50% green energy site.
- Designed a site-wide SCADA system to monitor and automate the site microgrid using Ignition as the front end.

2022.05 - 2022.08

### **SCADA Intern**

SpaceX

- Automated various power distribution tasks such as load-shedding, interactive fuel-consumption projector, and predictive site-load calculator that is still utilized during launch operations.
- Deployed numerous PLCs, control panels, and sensors to increase grid visibility and control using Allen Bradley, Siemens, and SEL hardware. Communication protocols used: MQTT, DNP, Modbus, Profinet, GOOSE.

2021.05 - 2021.08

### **Control Systems Intern II**

ICS Texas

- Designed and implemented control systems for numerous applications including natural gas sites, boiler systems, and production facilities. Primarily completed with Allen Bradley PLC hardware.

2020.05 - 2020.08

### **Control Systems Intern I**

ICS Texas

- Mastered AutoCAD for the design of control panels that were manufactured in-house. Developed an automated AutoCAD program in LISP to synthesize to accelerate manufacturing.

## Education

---

2023 - 2027

 College Station, TX

### **PhD**

Texas A&M University

*Electrical Engineering*

2019 - 2023

 College Station, TX

### **Bachelor of Science**

Texas A&M University

*Electrical Engineering*

## Leadership

---

2025.05 - PRESENT

### **Senator**

Graduate and Professional Student Government

*Student Senator for the ECEN Graduate Department*

- Draft legislation that upholds university values and improves the Aggie experience.
- Vote and represent the Department of Electrical Engineering on University policies and procedures.

## Publications

---

2026.02

### **Spectral Graph Filters for Large Power Systems**

In the proceedings of 2026 IEEE Texas Power and Energy Conference (TPEC)

2025.12

### **Using Spectral Graph Wavelets to Analyze Large Power System Oscillation Modes**

59th Hawaii International Conference on System Sciences (HICSS)

|            |                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025.01.01 | <b><u>Voltage Stability in Plausible Non-Uniform Geomagnetic Disturbances</u></b><br>IEEE Transactions on Power Systems                                           |
| 2025.01.01 | <b><u>EMT Simulation with Spectral Graph Wavelets</u></b><br>In the proceedings of 2025 IEEE Texas Power and Energy Conference (TPEC)                             |
| 2024.01.01 | <b><u>Voltage Stability Interface Limits in Non-Uniform Geomagnetic Disturbances</u></b><br>In the proceedings of 2024 56th North American Power Symposium (NAPS) |

## Presentations

|                                                                                                            |                                                                                    |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  College Station, Texas   | <b><u>Graduate Speaker</u></b><br>ECEN Townhall                                    |
|  College Station, Texas   | <b><u>Graduate Speaker</u></b><br>ECEN Fellowship Banquet                          |
|  College Station, Texas   | <b><u>Speaker</u></b><br>Modern Methods in Electromagnetic Transient Simulations   |
|  College Station, Texas | <b><u>Speaker</u></b><br>Voltage Stability in Non-Uniform Geomagnetic Disturbances |

## Awards

|            |                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2025.12.01 | <b><u>Best Paper</u></b><br>59th Hawaii International Conference on System Science<br><i>Best Paper in Electric Energy Systems Track</i> |
| 2025.11.01 | <b><u>Department Poster Award</u></b><br>Texas A&M University<br><i>2nd Place in Department Poster Event</i>                             |
| 2024.03.01 | <b><u>Michael Powell Graduate Fellowship</u></b><br>Texas A&M University                                                                 |
| 2024.03.01 | <b><u>Thomas Powell 1962 Fellowship</u></b><br>Texas A&M University                                                                      |

## Projects

|                   |                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 2025.12 - 2026.02 | <b><u>Blackout USA</u></b><br><i>An educational tool for all backgrounds to understand the challenges of operating the electric grid.</i> |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|